Minimizing exotic matter in rotating Ellis–Bronnikov wormholes

Nikita M. Shirokov[1, *] and Ilya Nachevsky[2, †]

[1] *Moscow Institute of Physics and Technology, Moscow, Russia*
[2] *Higher School of Economics, Moscow, Russia*

(Dated: October 31, 2026)

We ask whether a scalar potential can reduce the exotic matter needed to support a rotating Ellis–Bronnikov wormhole while keeping its throat size and angular momentum fixed. An automated search varies the potential and solves the stationary field equations for each trial. Additional Einstein constraints check the resulting geometry. For a selected slowly rotating drainhole, a calculation on a finer grid retains a 61% reduction in volume-integrated null-energy violation relative to the massless scalar solution. Comparisons of three selected drainholes show that the scalar variation becomes concentrated near the throat: the smaller volume integral accompanies stronger peak violation, stronger peak tides and longer finite-distance light transit. Three-dimensional evolutions initially show decreasing constraint errors as the grid is refined. The throat later contracts, with onset moving later at higher resolution. The potential therefore lowers the stationary matter requirement for this solution, while its subsequent evolution remains sensitive to numerical resolution.

I. INTRODUCTION

A traversable wormhole must violate the null energy condition (NEC) near its throat [3, 4]. The Ellis–Bronnikov solution provides a simple field model: a massless scalar with a reversed kinetic sign supports the throat [1, 2]. This phantom scalar supplies the required exotic matter, but the static wormhole is radially unstable [5, 6, 15, 16].

How much exotic matter is needed depends on both its distribution and the geometry. Visser, Kar, and Dadhich showed that the integrated violation can be made arbitrarily small in idealized thin-shell constructions [7, 8]. Our interest is in smooth scalar-field solutions that can also serve as initial data for nonlinear evolution. Rotation already reduces their exotic-matter integral along the families of Kleihaus and Kunz [11, 12, 16].

Those rotating solutions use a massless scalar without a potential. We instead allow a potential that changes how the scalar varies between the two ends of the wormhole. This can alter both the distribution of exotic matter and the geometry it supports. To isolate that effect, we compare solutions at the same equatorial throat radius and angular momentum.

Ordinary bosonic matter provides another way to modify the throat geometry. Dzhunushaliev et al. [10] constructed boson stars containing phantom-supported wormholes. Their solutions retain a radial unstable mode, although suitable boson self-interactions can bring its eigenvalue close to zero.

Here we vary the phantom potential, solve for the scalar and metric together, and measure the integrated NEC violation. The optimizer uses these measurements to choose further trials (Fig. 1). Independent Einstein constraints test whether each numerical solution is consistent with equations that were not used to construct it. Promising candidates are then checked on a finer grid before a selected solution is evolved in three dimensions.

Section II defines the model and the matter measure. Sections III and IV explain the search and evolution methods. Section V presents the search and evolution results, and Sec. VI examines the matter distribution and physical trade-offs, with numerical checks in Appendices A and B. We use units in which Newton's constant and the speed of light satisfy $G = c = 1$.

II. THE MATTER MODEL

We consider Einstein gravity coupled to the classical Ellis–Bronnikov phantom scalar ψ, augmented by a potential $\mathcal{V}(\psi)$,

$$S = \frac{1}{16\pi}\int d^4x\,\sqrt{-g}\,(R + 2\,g^{\mu\nu}\partial_\mu\psi\,\partial_\nu\psi + 2\,\mathcal{V}(\psi))\,, \quad (1)$$

where g is the determinant of the spacetime metric and R is its Ricci scalar. This extends the massless model of Refs. [11, 12, 16]. The reversed kinetic sign permits NEC violation.

Varying the action gives the coupled Einstein–Klein–Gordon equations,

$$R_{\mu\nu} = -2\,\partial_\mu\psi\,\partial_\nu\psi - \mathcal{V}\,g_{\mu\nu}, \qquad \Box\psi = \tfrac{1}{2}\mathcal{V}'(\psi). \quad (2)$$

Here $R_{\mu\nu}$ is the Ricci tensor, $\Box$ is the covariant wave operator, and a prime denotes a derivative with respect to ψ. Without a potential, the scalar has the analytic arctan profile of Ref. [12]. With a potential, both the scalar and the metric must respond. In particular, $\mathcal{V}$ supplies a source for the Einstein combination $R^t{}_t + R^\varphi{}_\varphi$. We therefore introduce an additional metric function B beyond the massless ansatz:

$$\begin{aligned} ds^2 = &-e^f dt^2 + B^2 e^{-f}\Big[e^\nu\left(d\eta^2 + h\,d\theta^2\right) \\ &+ h\sin^2\theta\,(d\varphi - \omega\,dt)^2\Big], \end{aligned} \quad (3)$$

* shirokov.nm@phystech.edu
† ommnammashivaya@gmail.com

with $h = \eta^2 + \eta_0^2$. The coordinates η and θ describe the radial and polar directions; η_0 is the throat parameter. The metric functions f, ω, ν and B, together with ψ, depend on both coordinates. The massless solutions have $B = 1$. We solve for the field value $\psi_- = \psi(-\infty)$ along with these functions.

The two families differ in the relation between their asymptotic ends. We set $f \to 0$ at $\eta \to +\infty$ and $f \to \gamma$ at $\eta \to -\infty$. The symmetric family has $\gamma = 0$, whereas the drainhole family has $\gamma = -\pi$ [16]. For the latter we use $\eta_0 = 2$, so its static massless member has unit mass.

The potential must vanish at both ends so that they remain asymptotically flat. Its derivative must also vanish at the limiting field values. We choose a double-well form with vacua at $\psi = 0$ and $\psi = \psi_-$:

$$\mathcal{V}(\psi) = \frac{m^2}{\psi_-^2}\,\psi^2(\psi - \psi_-)^2\,Q(u), \qquad u = \frac{\psi}{\psi_-}, \tag{4}$$

where $Q(u) = 1 + q_1 u + q_2 u^2 + q_3 u^3$. The prefactor fixes the two vacua, while the polynomial changes the barrier between them.

The search varies (m, q_1, q_2, q_3). The parameter m controls the curvature of the potential near the first vacuum, $\mathcal{V}''(0) = 2m^2$. The coefficients q_i change the height and asymmetry of the barrier. We require $Q(u) > 0$ between the vacua; individual coefficients may be negative. Setting $m = 0$ recovers the massless model.

To compare the matter content of different solutions, we integrate the radial NEC violation and normalize it by the equatorial throat radius R_e [7, 16]:

$$X = \frac{1}{R_e}\left|\int (\rho + p_\eta)\,dV\right|. \tag{5}$$

Here ρ is the energy density, p_η is the radial pressure, and dV is the proper spatial volume element. We call X the exoticity. The potential cancels directly from $\rho + p_\eta$, but it changes the scalar gradient and the volume element through the coupled field equations. Solving those equations is therefore essential to finding whether a trial potential lowers X.

We hold R_e and the angular momentum J at the values of a chosen massless reference solution. This makes the comparison about the potential's effect, rather than a change in throat size or spin.

III. OPTIMIZATION

The search alternates between constructing a wormhole and choosing a new potential. For each trial, a spectral Newton solver adjusts the scalar and geometry until the stationary field equations are satisfied numerically. The covariance-matrix adaptation evolution strategy (CMA-ES) [18] ranks the trials and uses them to guide the next proposals. Figure 1 shows the checks and measurements linking these steps.

A. Spectral Newton solver

The spectral representation covers both asymptotic ends on a finite computational domain. We use $x = (2/\pi)\arctan(\eta/\eta_0)$ to map them to $x = \pm 1$, and expand the fields in Chebyshev polynomials radially and even cosine modes angularly. A tensor-product Lobatto grid supplies the points at which the equations are evaluated.

Newton iteration computes a correction to the fields from the residual equations and their Jacobian. Damping reduces the correction when a full step would not improve the solution. We solve for f, ω, $\nu/\sin^2\theta$, B, and ψ; dividing out $\sin^2\theta$ regularizes the metric variable on the symmetry axis. The vacuum value ψ_- and a radial length scale are additional unknowns, with two equations enforcing the chosen R_e and J.

The Jacobian describes how the residual changes when the unknowns change. We compute it using automatic differentiation in `JAX` [19], then solve the resulting dense linear system on a graphics processing unit (GPU). Two independent Einstein constraints check the converged fields. We also inspect the highest spectral coefficients: a slowly decaying tail signals structure that the grid may not resolve. Appendix A gives the background checks.

B. CMA-ES

CMA-ES draws a population of trial potentials from a distribution in parameter space. After each population is evaluated, the better candidates guide the location and spread of subsequent proposals. For each trial, Newton iteration starts from the nearest previously solved potential, which supplies an initial guess for the fields.

The score combines exoticity with penalties for failed checks. Potentials that violate $Q > 0$, fail to converge, or exceed the background-constraint ceiling receive a large penalty. Smaller constraint violations receive a smaller penalty. The searches also used a radial perturbation screen of the static counterpart, described in Appendix A. The stationary results in this paper are selected by their exoticity and background constraints according to Sec. IV C.

The searches explore different regions of potential space for the two wormhole families (Fig. 2). Labels D and S denote drainhole and symmetric reference solutions; the suffix gives $100v_e$, where v_e is the reference equatorial rotation velocity. Each search keeps its own reference throat size and angular momentum fixed.

The parameter projections show which combinations were sampled, rather than an interpolated surface of solutions. Points can overlap even when the other two potential parameters differ. The progress curves divide each search's best value by its first candidate below the constraint ceiling, so their slopes show improvement within a search.

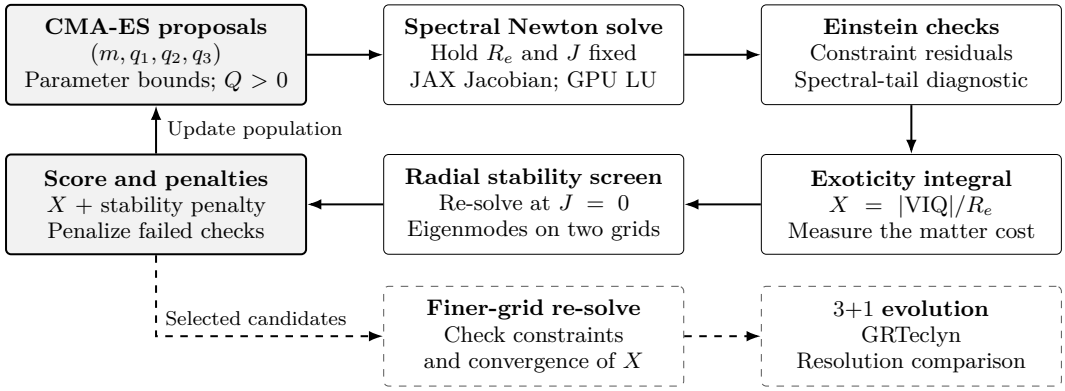


FIG. 1. Optimization of the confining potential at fixed throat size and angular momentum. Each CMA-ES proposal starts Newton iteration from a nearby solved candidate; JAX supplies the Jacobian for the GPU solve. Exoticity, the radial screen and constraint penalties feed back into the search; inadmissible or failed solves receive a penalty. Dashed arrows lead to separate resolution and evolution tests. The loop ranks candidates; the subsequent tests assess their reliability.

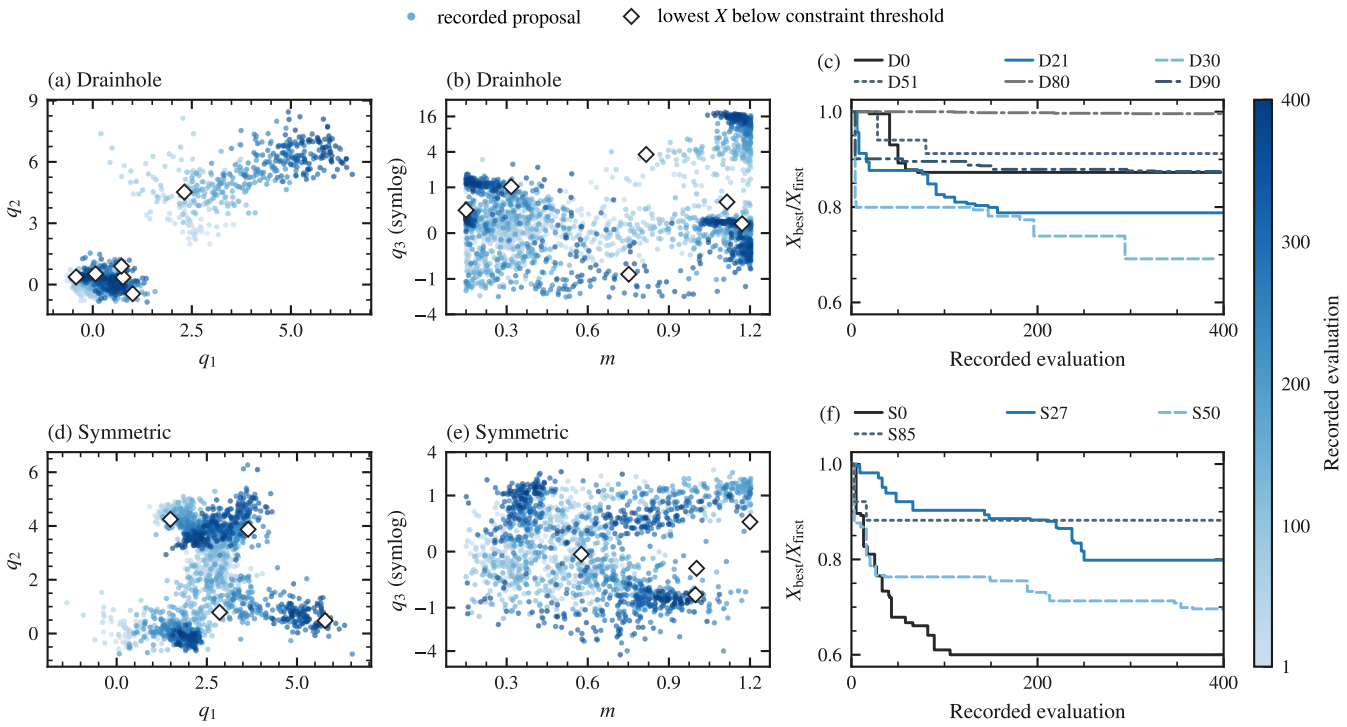


FIG. 2. CMA-ES exploration of the four potential parameters at fixed R_e and J, for the drainhole (top) and symmetric (bottom) families. Panels (a,b,d,e) project recorded proposals, coloured by evaluation order; diamonds mark each search's lowest X with both background-constraint residuals below 10^{-6}. The q_3 axes are symmetric-logarithmic, linear for $|q_3| \leq 1$. Panels (c,f) show the running minimum relative to the first candidate below that threshold. The searches concentrate in distinct parameter regions and approach plateaus at different rates.

IV. NUMERICAL SETUP

A. Interpolation onto the grid

The stationary solution supplies the initial metric and scalar field for evolution on a Cartesian grid with adaptive mesh refinement (AMR). We interpolate the background table using cubic splines with continuous second derivatives. The isotropic radius r is related to the stationary coordinate by $\eta = r - \eta_0^2/(4r)$, so the second asymptotic end is represented by a puncture at $r = 0$ [16].

The additional metric function B enters the conformal factor through $\chi \to \chi/B^2$. The lapse α, shift β^i, and scalar ψ are read from the stationary data. The extrinsic curvature K_{ij} follows from the stationary shift, and the scalar's initial conjugate momentum vanishes in the

corotating frame. The exported data also contain the potential coefficients needed during evolution.

B. Evolution scheme

We evolve these data with `GRTeclyn` [23–25], following the formulation used in Refs. [15, 16]. The metric evolution uses the conformal and covariant Z4 (CCZ4) equations [20], with 1+log slicing [21] and the Gamma-driver shift [22]. Spatial derivatives and Runge–Kutta time integration are fourth order, with Kreiss–Oliger dissipation.

The scalar evolution includes the potential and its derivative. The Hamiltonian constraint diagnostic includes the potential contribution as well. Mesh refinement concentrates resolution near the throat. The optional outer sponge term that drives fields toward their reference values is disabled. Appendix B specifies the grid and timestep used for the resolution study.

C. Criteria

A low score identifies a promising trial, but the reported stationary candidates must also satisfy the independent background equations. We distinguish the loose constraint ceiling used during exploration from the stricter threshold used for Tables I and II. The former is 10^{-6}; the latter is 10^{-8} for each of the two residuals. Trials between these thresholds receive a constraint penalty.

The tables select the lowest recorded X meeting the stricter threshold in each search. A separate row identifies the D21 candidate re-solved on a finer grid and used for evolution. Perturbative stability requires an additional test: a converged mode must satisfy the independent linearized Einstein constraints as well as its eigenvalue equations.

V. RESULTS

A. Stationary exoticity at fixed throat and spin

The search finds lower exoticity at most of the sampled rotations in both families. Tables I and II give the potential parameters and the corresponding matter integrals, using the same model order. Their search-grid rows describe the minima found at that numerical resolution. Figure 3 places these candidates relative to the massless families.

The reduction is not uniform across spin. In particular, the high-spin D90 search candidate remains above its massless counterpart. The tested potential family and search therefore do not improve on every reference solution.

For the selected D21⋆ solution, the finer-grid calculation retains a reduction of about 61% in exoticity at the same throat radius and angular momentum. Its background residuals satisfy the independent constraints (Appendix A). This is the solution used for the three-dimensional evolution study; it is distinct from the lowest point in the search history.

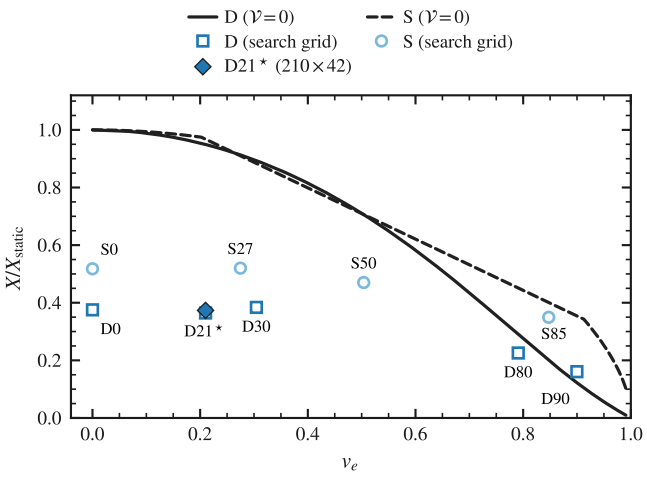


FIG. 3. Exoticity relative to each family's massless static member, against the reference rotation v_e. Black curves: massless families. Open blue markers: the search-grid candidates of Table II. The filled diamond identifies D21⋆ on 210×42. Most search candidates lie below their massless counterparts; D90 lies above its counterpart.

B. Three-dimensional evolution of D21⋆

We evolved D21⋆ on three grids to examine how the dynamics change with resolution. The domain and refinement hierarchy were held fixed while the grid spacing decreased. Early in the evolution, the constraint errors near the throat fall approximately at fourth order. Other refinement levels show more variable behavior, as quantified in Appendix B.

The throat later contracts on all three grids, with contraction starting later as resolution increases. Figure 6 shows this shift in the radius histories alongside the wave and lapse diagnostics. Thus the early constraint check and the later dynamics have different numerical behavior: the former shows decreasing error, while the contraction onset is still sensitive to grid spacing.

At $N = 128$, the resolution used for classical D21 in Ref. [16], D21⋆ contracts later and has a lower measured growth rate (Fig. 4). We fit $\ln|dR_A/dt|$ over the same 0.2–1% areal-radius decrease. The growth time increases from $\tau = 6.59$ to 9.49, corresponding to a 31% lower rate. The caption records the differences in background and outer damping settings. A static potential-bearing D0 solution also has a lower fitted growth rate, by 30%, with the reference throat size and outer damping held fixed.

TABLE I. Potential parameters of the lowest-X candidates with both background-constraint residuals at most 10^{-8} on the 150×30 search grid. D and S denote the drainhole and symmetric families. D21* is the separately selected solution re-solved on 210×42, rather than the search-grid minimum.

model	m	q_1	q_2	q_3
Drainhole, 150×30 search grid				
D0	1.199	0.227	0.485	0.274
D21	1.171	0.762	0.351	0.206
D30	0.740	0.607	0.201	−0.664
D80	0.234	0.083	0.504	1.187
D90	0.150	−0.337	0.366	0.445
Symmetric, 150×30 search grid				
S0	1.198	2.433	−0.046	1.197
S27	0.855	3.232	3.776	−0.256
S50	0.962	3.739	0.922	−0.828
S85	0.425	2.438	4.267	0.445
Selected 210×42 solution				
D21*	1.023	0.672	0.048	0.310

TABLE II. Exoticity of the same candidates as Table I, compared with the massless solution at the same R_e and J. Here $\Delta X = 100(X/X_{\mathcal{V}=0} - 1)$. The search-grid rows and the selected finer-grid solution are listed separately.

model	X	$X_{\mathcal{V}=0}$	ΔX
Drainhole, 150×30 search grid			
D0	1.839	4.898	−62.5%
D21	1.785	4.645	−61.6%
D30	1.880	4.376	−57.0%
D80	1.105	1.419	−22.2%
D90	0.787	0.600	+31.3%
Symmetric, 150×30 search grid			
S0	1.627	3.142	−48.2%
S27	1.634	2.994	−45.4%
S50	1.477	2.625	−43.7%
S85	1.098	1.462	−24.9%
Selected 210×42 solution			
D21*	1.829	4.645	−60.6%

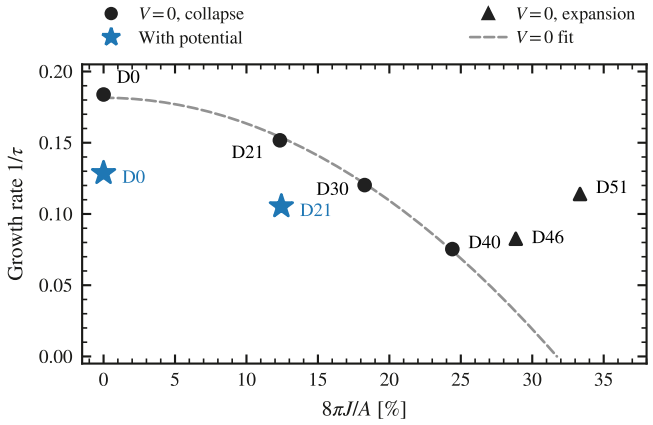


FIG. 4. Measured growth rates at $N = 128$: classical models from Ref. [16] and potential-bearing D0 and D21* (blue stars). The dashed curve fits the classical contracting models. Classical D21 has $J = 1$ and outer drive strength 1; D21* has $J = 1.0138$ and drive strength 0, with initial equatorial radii differing by about 0.1%. D0 uses the same reference throat size and drive strength 0 in both runs. Over the same 0.2–1% areal-radius decrease, both potential-bearing models have about 30% lower fitted growth rates.

VI. PHYSICS OF THE OPTIMIZED SOLUTIONS

The potential lowers the volume-integrated exoticity by concentrating the scalar variation into a narrower region near the throat. It does not weaken the NEC violation everywhere. We examine this redistribution in three selected 210×42 drainholes, denoted D0*, D21* and D30*. These are finer-grid selections, distinct from the search minima in Tables I and II; D21* is the solution already identified there. Each has its own massless reference at the same R_e, J and asymmetry. In particular, both members of the stationary D21 comparison have $J = 1.0138$.

A. Matter distribution and throat geometry

Figure 5 compares profiles against physical proper distance, rather than the radial coordinates of different numerical solutions. The local radial NEC deficit is the squared orthonormal scalar gradient divided by 4π. Here the radial null vector has unit energy in the local zero-angular-momentum observer's frame. The potential cancels from its direct null contraction, but changes the scalar and metric through the field equations.

In D21*, the scalar excursion between the two ends falls from 3.46 to 1.25. Its remaining variation occurs over a much shorter distance, so the peak radial NEC violation is 3.4 times stronger. The proper width between the 5% and 95% cumulative contributions to X decreases from $27.16R_e$ to $0.65R_e$. This measure includes the long classical tails beyond the plotted interval; it describes the matter distribution, not a change in throat radius. Angular scalar gradients remain small, giving a narrow radial layer rather than an equatorial ring.

The same pattern occurs in the other two solutions (Table III). Their narrower distributions outweigh the higher peaks, reducing X by 56–62% across the three models. The equatorial throat radius is fixed, and the throat area changes by at most 0.6%. The rotating throats become slightly less oblate. Thus the reduction mainly reflects a changed scalar distribution, while the overall throat dimensions remain similar.

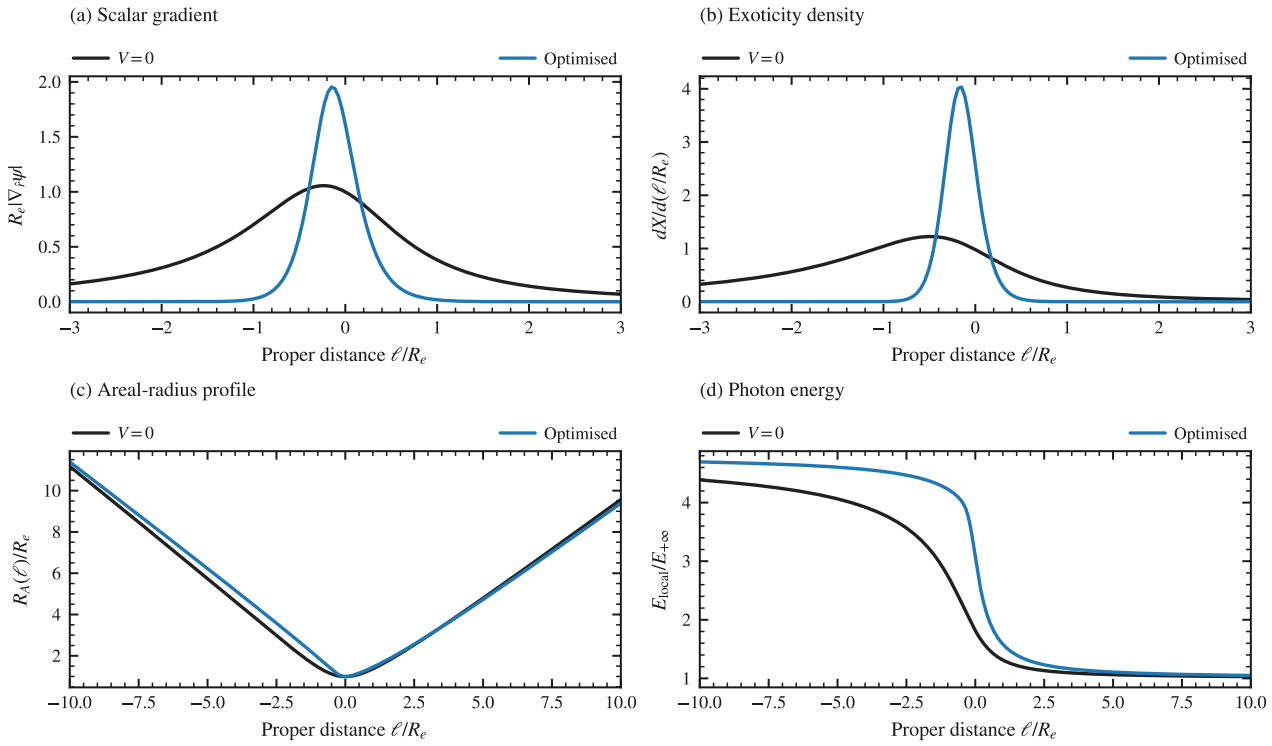


FIG. 5. Stationary profiles of D21* and its massless reference at the same R_e and J. The signed equatorial proper distance ℓ is zero at the minimum-area throat. Panels show the equatorial scalar gradient, the angularly integrated contribution to X, the area radius of constant-η surfaces, and the energy of a zero-angular-momentum photon measured by a local observer relative to its conserved energy at positive infinity. The potential concentrates the scalar gradient and exoticity near the throat while sharpening the interior redshift profile.

TABLE III. Stationary quantities divided by their own massless-reference values at fixed R_e, J and asymmetry. Stars identify the selected finer-grid solutions. The width includes the complete radial distribution; the null NEC integral follows the complete equatorial ray with unit Killing energy at positive infinity. Peak tides and light transit use equatorial proper distances $-10R_e$ to $+10R_e$. A smaller volume integral accompanies stronger local violation, stronger peak tides and longer transit.

Quantity	D0*	D21*	D30*
Total exoticity X	0.376	0.394	0.439
Central-90% proper width	0.0204	0.0238	0.0377
Peak radial NEC violation	3.75	3.43	2.46
Ray-integrated NEC magnitude	1.42	1.39	1.33
Peak tides, $v = 0.5$	6.57	5.93	3.87
Light transit time	1.180	1.167	1.146

B. Tides, redshift and light propagation

A smaller volume integral does not imply a gentler passage through the throat. We measure tides by the largest absolute eigenvalue of the electric Riemann tensor in a radially moving observer's frame. At the same local speed $v = 0.5$, its peak over the matched interval increases in all three solutions. This compares local forces along a fixed-speed scan; a freely falling traveller's accumulated tidal exposure also depends on its trajectory. The stronger peak is consistent with the sharper scalar and metric variation visible in Fig. 5.

The asymptotic redshift remains fixed by the common asymmetry γ, while the interior profile changes. For a zero-angular-momentum photon, the local energy is $E_{\rm local}/E_{+\infty} = 1/\alpha$, where $\alpha = e^{f/2}$ is the lapse and $E_{+\infty}$ is the conserved energy normalized at positive infinity. The optimized profile makes the transition between the two ends more abrupt.

We compute a finite light-crossing time along the equatorial null geodesic by integrating $dt/d\ell = 1/\alpha$ between endpoints at equal proper distances from each throat. The same positive-infinity clock normalizes t in both solutions. Transit is longer for the optimized solutions (Table III); the increase also occurs when the endpoint distances are halved or doubled, although the ratio changes. These are wormhole-to-wormhole comparisons at specified endpoints.

The magnitude of the NEC integral along the complete equatorial null ray also increases. Its affine parameter is normalized by unit Killing energy at positive infinity, unlike the local normalization used for the peak NEC

deficit. The ray and volume integrals therefore measure different aspects of the same distribution. Minimizing X reduces the volume-weighted matter requirement while concentrating the violation encountered near the throat.

VII. CONCLUSIONS

Adding a scalar potential can lower the stationary exotic-matter requirement without changing the reference throat size or angular momentum. The search combines numerical solutions of the field equations with independent Einstein constraints, and it retains a reduction of about 61% for the selected D21 solution on a finer grid. The benefit depends on the reference solution; the high-spin D90 search does not improve on its massless counterpart.

The D21 evolutions initially show decreasing constraint errors under refinement and later develop contraction. Its onset moves later on finer grids.

The stationary comparisons explain the exoticity reduction: the scalar variation is confined to a narrower region near the throat, with little change in the throat's overall dimensions. The local violation becomes stronger, and the selected solutions have larger peak tides and longer finite-distance light transit. Reducing the volume-integrated matter requirement therefore carries a trade-off in these measures of passage through the wormhole.

VIII. REPRODUCIBILITY

The solver, optimization and figure-generation scripts are available in the `GRTeclyn` research repository [26]. The archived search traces identify every candidate used in Tables I and II; a checked selection ledger generates both tables. A separate checked ledger supplies Table III from the saved D0*, D21* and D30* fields. Those fields, their massless references, physical profiles and numerical quadrature checks are archived alongside the final evolution diagnostic streams. All article figures and tables rebuild from those tracked data.

ACKNOWLEDGMENTS

This work was supported by Gravity Frontiers (https://www.gravityfrontiers.org/en), which funded the underlying research and the development of the numerical-simulation software. The authors acknowledge the GRTL Collaboration for `GRChombo`, `GRTresna`, and `GRTeclyn`.

Appendix A: Solver and stability checks

The massless limit provides a check against the established wormhole solutions. Setting $\mathcal{V} = 0$ reduces the spectral equations to the system used in Ref. [16]. Newton iteration reproduces the saved solutions' global charges, $B = 1$, and exoticity values to machine precision.

For nonzero potential, we use two Einstein equations that do not enter the Newton solve: the $\eta\theta$ component and the generalization of the scalar-charge identity of Ref. [12]. Their residuals, denoted E_{19} and E_c, assess the consistency of the computed background independently of the equations solved. For D21* on 210×42, they are $E_{19} = 1.6 \times 10^{-10}$ and $E_c = 4.3 \times 10^{-10}$, below the selection threshold of Sec. IV C.

For the stationary diagnostics of Sec. VI, doubling the radial quadrature and profile sampling, tightening null-ray integration, and varying the angular quadrature leave the quoted values unchanged at the displayed precision. Exact Ellis limits test the gradient, null NEC, transit and tidal normalizations. An independent proper-distance calculation of the spherical tidal tensor also agrees with the D0* measurement.

The radial screen examines small spherically symmetric perturbations of the static counterpart at the same throat radius. Linearizing the Einstein–Klein–Gordon equations gives three coupled radial wave equations. On a Chebyshev–Lobatto grid these become a generalized eigenvalue problem.

A numerical eigenvalue alone does not identify a physical mode, because gauge modes can also appear in the discretized problem. A physical mode must agree between two grids and have small residuals in the linearized tt and $t\eta$ Einstein constraints. On classical massless backgrounds, the calculation reproduces the published dimensionless growth rates $|\omega_I|\eta_0 = 0.390$ for the drainhole [13, 14] and 1.182 for the symmetric Ellis throat. Here $|\omega_I|$ is the exponential growth rate. The stationary candidate selection uses the background constraints specified above.

Appendix B: Convergence of the D21 evolution

The resolution study used base grids with $N = 96, 128, 192$ points across a domain of length $L = 64$. Three additional AMR levels refine the throat region, and the timestep is $\Delta t = 0.02\,\Delta x$. The runs use the same potential and numerical settings; only the base resolution changes.

Figure 6 compares the constraint norms over the common early interval through $t = 50$. The lower panels show the later diagnostic histories. The areal radius is estimated from the minimum area of coordinate spheres. The wave diagnostic is the change in $r\,\mathrm{Re}\,\Psi_4^{20}$ at extraction radius $r = 14$, where Ψ_4^{20} is the quadrupolar, axisymmetric mode of the Newman–Penrose curvature scalar.

Refining the grid should reduce the constraint errors, since the continuum constraints vanish. To measure how quickly they decrease, we compare norms at the same time and refinement level. For two base resolutions N_a

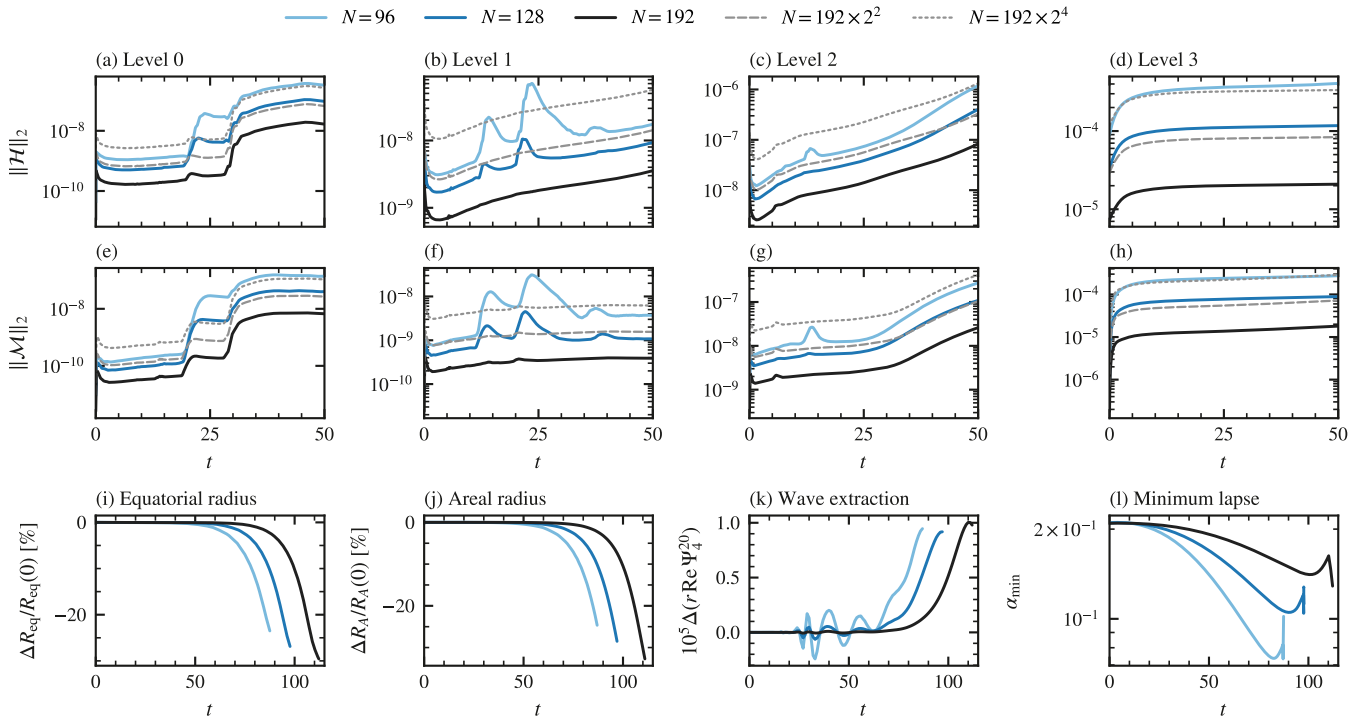


FIG. 6. Resolution study of D21* at $N = 96, 128, 192$. Top two rows: Hamiltonian and momentum constraint norms through $t = 50$; grey dashed and dotted curves multiply the finest-grid norms by 2^2 and 2^4. Bottom row: fractional equatorial- and areal-radius changes, the change in $r\,\mathrm{Re}\,\Psi_4^{20}$ at $r = 14$, and minimum lapse over each recorded history. Near-throat constraints scale approximately at fourth order early on; the later contraction begins at resolution-dependent times.

TABLE IV. Constraint orders from Eq. (B1), using linearly interpolated norms. Each cell gives 96 → 128 / 128 → 192 at the listed time and level. Finest-level orders stay near four; coarser levels vary.

constraint	$t = 10$	$t = 20$	$t = 30$	$t = 40$	$t = 50$
Ham L0	2.67 / 2.77	1.09 / 3.13	4.66 / 4.35	4.39 / 4.27	4.28 / 4.30
Mom L0	2.45 / 2.60	0.47 / 3.25	4.98 / 4.84	4.42 / 4.46	4.32 / 4.41
Ham L1	2.12 / 2.38	2.80 / 2.49	3.95 / 2.46	2.33 / 2.47	2.15 / 2.39
Mom L1	2.00 / 2.33	4.99 / 3.30	7.08 / 2.71	3.70 / 3.02	4.28 / 2.51
Ham L2	2.17 / 2.43	2.21 / 2.49	2.35 / 2.50	3.45 / 3.12	3.78 / 3.90
Mom L2	2.39 / 2.47	2.23 / 2.50	2.62 / 2.94	3.19 / 3.63	3.19 / 3.53
Ham L3	4.03 / 4.18	4.10 / 4.21	4.14 / 4.23	4.20 / 4.24	4.33 / 4.25
Mom L3	3.93 / 4.26	3.94 / 4.28	3.91 / 4.24	3.86 / 4.11	3.93 / 3.93

and N_b, the measured order is

$$p_{a\to b}(t) = \frac{\ln[C_{N_a}(t)/C_{N_b}(t)]}{\ln(N_b/N_a)}, \tag{B1}$$

where C_N is the Hamiltonian or momentum constraint norm. An order near four means that halving the grid spacing reduces the error by roughly a factor of sixteen. Table IV gives both neighboring-resolution comparisons. The finest level stays close to this behavior, while intermediate levels show mixed orders and transients.

The radius histories give a separate test of the dynamics. Using a one-percent decrease in equatorial radius to mark contraction, the first recorded crossings occur at $t = 58.3$, 67.3, and 80.7 in order of increasing resolution.

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